

# A claimed Schatten triangle constant smaller than one

## Disproof of conjecture 00000008574

ziangni-sys

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### The general- $p$ clause

Both language versions assert that the triangle constant for the Schatten quasi-norm is

$$C_p = 2^{1-1/p} \quad \text{for general } p.$$

The Schatten classes and their quasi-norms are defined for  $p > 0$ , including  $p = 1/2$ . We disprove this clause at that positive value. The source gives no restriction excluding  $p < 1$ . The assertions at  $p = 1$ , about rank-one attainment, and about the large- $p$  limit are not needed for the disproof.

By triangle constant we mean a coefficient  $C$  for which

$$\|A + B\|_{S_p} \leq C(\|A\|_{S_p} + \|B\|_{S_p})$$

holds for all operators in the class. A proposed exact optimal constant must, in particular, satisfy this inequality. We show that the displayed coefficient does not even do that.

### Actual one-dimensional operators and singular values

Take the complex Hilbert space  $H = \mathbb{C}$  with its standard orthonormal basis. Its operators have  $1 \times 1$  complex matrices. For  $M = [z]$ , the actual adjoint and Gram matrix are

$$M^* = [\bar{z}], \quad M^*M = [\bar{z}z] = [|z|^2].$$

A  $1 \times 1$  matrix  $[w]$  has exactly one eigenvalue,  $w$ : if  $wv = \mu v$  with  $v \neq 0$ , cancellation gives  $\mu = w$ , and conversely the vector  $v = 1$  is an eigenvector. Thus the sole singular value of  $M$  is the nonnegative square root of its sole Gram eigenvalue,

$$s(M) = \sqrt{|z|^2} = |z|.$$

These identities concern actual conjugate transpose, matrix multiplication and nonzero eigenvectors; singular-value numbers are not postulated.

The usual finite-dimensional Schatten formula is

$$\|M\|_{S_p} = \left( \sum_{i=1}^{\dim H} s_i(M)^p \right)^{1/p}.$$

Here the sum has exactly one term. Every such finite-dimensional operator belongs to  $S_p$  for all  $p > 0$ .

## The counterexample

Let  $A = B = I = [1]$ . These are actual rank-one identity operators on  $H$ ;  $A + B = 2I = [2]$ . Their Gram matrices are  $I, I, 4I$ , so their singular values are respectively 1, 1, 2.

At  $p = 1/2$ , for every nonnegative singular value  $s$  the Schatten expression is

$$(s^{1/2})^2 = s.$$

Consequently

$$\|A\|_{S_{1/2}} = 1, \quad \|B\|_{S_{1/2}} = 1, \quad \|A + B\|_{S_{1/2}} = 2.$$

The claimed coefficient, using actual real powers, equals

$$C_{1/2} = 2^{1-1/(1/2)} = 2^{-1} = \frac{1}{2}.$$

Its proposed triangle estimate becomes

$$2 \leq \frac{1}{2}(1 + 1) = 1,$$

which is false. This proves that  $2^{1-1/p}$  cannot be the general Schatten quasi-norm triangle constant. The same obstruction persists when the one-dimensional operators are embedded as rank-one operators in any larger Hilbert space.

## Formal coverage

The Lean project uses actual `Matrix (Fin 1) (Fin 1) Complex` values. It defines the Gram matrix by actual conjugate transpose and matrix multiplication. It proves the complete eigenvalue characterization for *every*  $1 \times 1$  complex matrix through actual matrix-vector multiplication and a nonzero vector. It proves that the Gram entry is the actual complex squared modulus, and that its nonnegative square root squares to the Gram eigenvalue.

The singular-value array has one entry because the dimension is one. The project defines the standard finite-dimensional Schatten formula using that array, a genuine finite sum, and Mathlib's real powers. It proves the half-exponent specialization using the square-root identity for real powers and the nonnegativity of the singular value. The identity and doubled-identity singular values are computed from their actual Gram matrices. No norms, eigenvalues, singular values, or claimed constants are assumed as data.

The final theorem negates the universal triangle-bound predicate for all  $1 \times 1$  complex matrices at  $p = 1/2$  and the source's coefficient. Any universal Schatten triangle estimate would imply this finite-dimensional specialization, so the theorem supplies a complete counterexample to the general claim.

## Verification

Lean 4.19.0 with a pinned Mathlib v4.19.0 revision is used. Full project build and complete-source check logs include the theorem axiom audits. All identities and inequalities are kernel checked; no external numerical calculation, admitted proof, extra axiom or native decision short-cut is used. No auxiliary code is required. The compiled two-page PDF is visually inspected in full.